

Aaquib Tabrez

Curriculum Vitae

Postdoctoral Scholar
University of Southern California
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Research Interests

My research objective is to develop autonomous agents that communicate, coordinate, and collaborate with humans to enhance human capabilities through mutual understanding, enabling safe, explainable, and trustworthy human-robot teaming. My work focuses on: 1) Establishing shared understanding between humans and agents by aligning mental models, capabilities, and values for effective collaboration; 2) Enabling multimodal communication so that humans can learn from agents, and agents can adapt to humans, through natural, interactive exchanges; 3) Designing explainable agents capable of communicating their intentions, limitations, and learned knowledge to foster trust and enable safe, transparent real-world deployment.

I ground my research in interactive tasks that involve physical and virtual robots, and draw upon methods from human-robot interaction, explainable AI, reinforcement learning, and foundation models.

Professional Experience

2024 – **University of Southern California**, Postdoctoral Scholar.

present Advisor: Prof. Stefanos Nikolaidis

2014 – 2016 **Daimler, Chennai, India.**

Procurement Manager

Education

2019 – 2024 **University of Colorado, Boulder**, GPA: 4.0, Ph.D., *Computer Science*.

Advisor: Prof. Bradley Hayes

Dissertation: *Mediating Trust and Influence in Human-Robot Teams via Multimodal Communication and Explanation for Mental Model Alignment* ([Link to Dissertation](#))

Committee: Bradley Hayes*, Alessandro Roncone, Daniel Szafir, Nisar Ahmed, & Sonia Chernova

2017–2019 **University of Colorado, Boulder**, GPA: 4.0, MS, *Mechanical Engineering*.

2010–2014 **National Institute of Technology Karnataka, India**, B.Tech, *Mechanical Engineering*.

Awards and Recognition

2024 **People's Choice Award - 3MT Competition.**

Won the People's Choice Award at the 2024 CU Boulder Three Minute Thesis (3MT) competition, receiving the highest number of audience votes.

2023 **Annual Research Expo'23 Poster Presentation Award.**

Received best research poster presentation award at the CU Boulder's Annual Research Expo '23.

2022 **Robotics: Science and Systems (RSS) Pioneers.**

Selected for workshop bringing together top early career researchers in robotics.

2022 **Best Student Paper Award Runner-up at AAMAS.**

For the paper "Descriptive and Prescriptive Visual Guidance to Improve Shared Situational Awareness in Human-Robot Teaming".

2022 **Won Spring 2022 Annual Research Expo Event.**

Received the best poster presentation award at the Spring 2022 Annual Research Expo from CU Boulder.

2020 **IBM PhD Fellowship Finalist.**

One of three students nominated by the CS department at CU Boulder.

- 2019 **Best Paper Award Finalist for Technical Advances at [ACM/IEEE HRI](#).**
For the paper "Explanation-based Reward Coaching to Improve Human Performance via Reinforcement Learning".
- 2019 **Human-Robot Interaction (HRI) [Pioneers](#).**
Selected for workshop bringing together top early career researchers in HRI.
- 2019 **[Awatar and Teji Singh Graduate Fellowship](#).**
A \$5,000 fellowship for early career PhD students demonstrating a strong academic and research record.

Papers in Submission

Asterisk (*) denotes shared first authorship

- [paper link](#) **Algorithmic Prompt Generation for Diverse Human-like Teaming and Communication with Large Language Models.**
Siddharth Srikanth, Varun Bhatt, Boshen Zhang, Werner Hager, Charles Michael Lewis, Katia P. Sycara, **Aaquib Tabrez**, Stefanos Nikolaidis
- [no paper link](#) **Title omitted for blind review.**
Aaquib Tabrez, Ryan Leonard, Bradley Hayes.

Journal Articles

- [paper link](#) **A survey of Mental Modeling Techniques in Human-Robot Teaming.**
Aaquib Tabrez, Matthew B. Luebbbers, Bradley Hayes.
Springer-Nature Current Robotics Reports, 2020

Conference Publications

- [paper link](#) **Recency Bias in Task Performance History Affects Perceptions of Robot Competence and Trustworthiness.**
Matthew B. Luebbbers*, **Aaquib Tabrez***, Kanaka Samagna Talanki, Bradley Hayes.
To appear: Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2024
- [paper link](#) **Autonomous Justification for Enabling Explainable Decision Support in Human-Robot Teaming.**
Aaquib Tabrez*, Matthew B. Luebbbers*, Kyler Ruvane*, and Bradley Hayes.
Robotics: Science and Systems (RSS), 2023
- [paper link](#) **Descriptive and Prescriptive Visual Guidance to Improve Shared Situational Awareness in Human-Robot Teaming.**
Aaquib Tabrez*, Matthew B. Luebbbers*, Bradley Hayes.
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2022
[Best Student Paper Runner-up \(Top 2 of 629 submissions\).](#)
- [paper link](#) **Asking the Right Questions: Facilitating Semantic Constraint Specification for Robot Skill Learning and Repair.**
Aaquib Tabrez*, Jack Kawell*, Bradley Hayes.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021
- [paper link](#) **Explanation-based Reward Coaching to Improve Human Performance via Reinforcement Learning.**
Aaquib Tabrez, Shivendra Agrawal, Bradley Hayes.
ACM/IEEE International Conference on Human Robot Interaction (HRI), 2019
[Best Technical Paper Runner-up.](#)

Workshop Publications

- [paper link](#) **Hierarchical Multi-Agent Reinforcement Learning with Explainable Decision Support for Human-Robot Teams.**
Aaquib Tabrez*, Matthew B. Luebbbers*, Kyler Ruvane*, Ashley H. Rabin, Kevin W. King, William Gerichs, and Bradley Hayes.
Proceedings of the Workshop on Explainability for Human-Robot Collaboration (X-HRI), 2024
- [paper link](#) **Autonomous Policy Explanations for Effective Human-Machine Teaming.**
Aaquib Tabrez.
Doctoral Consortium at the AAAI Conference on Artificial Intelligence, 2024
- [paper link](#) **Effective Human-Machine Teaming through Communicative Autonomous Agents that Explain, Coach, and Convince.**
Aaquib Tabrez
Doctoral Consortium at the International Conference on Autonomous Agents and Multiagent Systems, 2023
- [paper link](#) **Augmented Reality and Proxy Grippers Improve Demonstration-based Robot Skill Learning.**
Carl L. Mueller, Matthew B. Luebbbers, **Aaquib Tabrez**, and Bradley Hayes.
Proceedings of the Workshop on Life-Long Learning with Human Help (L3H2), 2023
- [paper link](#) **Mediating Trust and Influence in Human-Robot Interaction via Explainable AI.**
Aaquib Tabrez, Bradley Hayes.
Pioneers Workshop at Robotics: Science and Systems (RSS), 2022
- [paper link](#) **Augmented Reality-Based Explainable AI Strategies for Establishing Appropriate Reliance and Trust in Human-Robot Teaming.**
Matthew B. Luebbbers*, **Aaquib Tabrez***, Bradley Hayes.
Workshop on Virtual, Augmented and Mixed Reality for Human-Robot Interaction (VAM-HRI), 2022
- [paper link](#) **Solutions for Socially Intelligent HRI in Real-World Scenarios (SSIR-HRI).**
Karen Tatarian, Sera Buyukgoz, Marine Chamoux, **Aaquib Tabrez**, Bradley Hayes, Mohamed Chetouani.
Companion of the ACM/IEEE International Conference on Human-Robot Interaction, 2021
- [paper link](#) **Interactive Constrained Learning from Demonstration Using Visual Robot Behavior Counterfactuals.**
Carl Mueller, **Aaquib Tabrez**, Bradley Hayes
Workshop on Accessibility of Robot Programming and Work of the Future at RSS, 2021
- [paper link](#) **Emerging Autonomy Solutions for Human and Robotic Deep Space Exploration.**
Matthew B. Luebbbers*, Christine T. Chang*, **Aaquib Tabrez***, Jordan Dixon*, Bradley Hayes.
SpaceCHI: Human-Computer Interaction for Space Exploration, 2021
- [paper link](#) **Automated Failure-Mode Clustering and Labeling for Informed Car-To-Driver Handover in Autonomous Vehicles.**
Aaquib Tabrez*, Matthew B. Luebbbers*, Bradley Hayes.
Workshop on Assessing, Explaining, and Conveying Robot Proficiency for Human-Robot Teaming, 2020
- [paper link](#) **Improving human-robot interaction through explainable reinforcement learning.**
Aaquib Tabrez, Bradley Hayes.
Companion of the ACM/IEEE International Conference on Human-Robot Interaction, 2019

Academic Service, Memberships, and Public Outreach

- **Grant Review:** NSF Panelist (2025)
- **Workshop & Committee Leadership:**
 - Program Chair: RSS Pioneers (RSS 2023), HRI Pioneers (HRI 2021, HRI 2020)
 - Co-Organizer: Workshops at RO-MAN 2020/HRI 2021 (Social HRI) and RO-MAN 2023 (Explainability).
- **Conference Reviewer:** HRI, ICRA, IROS, RO-MAN, CHI, XAIP@ICAPS, XAI@IJCAI
- **Journal Reviewer:** RSS, IEEE RA-L, ACM T-HRI, IJHCI